



WEEK 01 ASSIGNMENT

Cybersecurity

Fundamentals &

PREPARED BY	Ahmed Mustafa
DOMAIN	Red Team Cybersecurity
REGISTRATION NO.	RED B01-5199
SUBMISSION DATE	4/July/2026

1. Introduction

- Cybersecurity is the practice of protecting computer systems, networks, applications, and digital data from unauthorized access, cyberattacks, and data breaches.
- With the rapid growth of digital technologies, organizations rely heavily on secure information systems to protect sensitive business and customer information.
- Modern cyber threats include malware, ransomware, phishing attacks, insider threats, data breaches, and denial-of-service (DoS/DDoS) attacks.
- The primary goal of cybersecurity is to ensure the **Confidentiality, Integrity, and Availability (CIA Triad)** of information and information systems.
- **Confidentiality** ensures that sensitive information is accessible only to authorized users.
- **Integrity** guarantees that data remains accurate, complete, and protected from unauthorized modification.
- **Availability** ensures that systems, applications, and data remain accessible to authorized users whenever needed.
- Organizations implement multiple security controls such as firewalls, antivirus software, encryption, intrusion detection systems (IDS), intrusion prevention systems (IPS), Virtual Private Networks (VPNs), and Multi-Factor Authentication (MFA) to strengthen security.
- Regular vulnerability assessments, penetration testing, security monitoring, and software updates help identify and eliminate security weaknesses before attackers can exploit them.
- Employee awareness and cybersecurity training play an important role in reducing human-related security risks such as phishing and social engineering attacks.
- Cybersecurity follows a **Defense-in-Depth** approach, where multiple layers of security work together to provide comprehensive protection against cyber threats.
- A strong cybersecurity framework not only prevents attacks but also enables organizations to detect, respond to, and recover quickly from security incidents.
- This report examines real-world cyberattacks, explains the CIA Triad, discusses commonly used cybersecurity tools, presents a secure home network topology, and describes the key responsibilities of a Cybersecurity Analyst.

2. Real-World Cyberattacks & CIA Triad Analysis

The following matrices outline five distinct historical cyber incidents, detailing their primary threat mechanisms, target surfaces, operational downfalls, and their specific violations of the core security vectors within the Confidentiality, Integrity, and Availability (CIA) Triad framework.

Incident & Year	Attack Vectors	Technical Description	Organizational Consequences
Equifax Breach (2017)	WebApplication Vulnerability	Intruders weaponized an unpatched, public remote code execution vulnerability within the Apache Struts server framework (CVE-2017-5638) to execute commands and navigate laterally into core consumer databases.	Compromised highly regulated personal identity records of 147 million consumers, leading to massive data exposure liabilities and statutory financial penalties.
Colonial Pipeline (2021)	Credential Compromise	Malicious actors leveraged exposed credentials on a legacy virtual private network profile lacking multi-factor authentication to plant malicious payloads and encrypt active network assets.	Paralyzed enterprise billing operations, forcing a complete operational shutdown of critical physical pipelines and creating severe regional fuel supply deficits.
SolarWinds Incident (2020)	Advanced Supply Chain Compromise	State-sponsored entities compromised a developer pipeline, inserting a covert backdoor module named SUNBURST into official, cryptographically verified build templates distributed to clients.	Exposed thousands of public sector and enterprise platforms to continuous lateral surveillance through routine, trusted vendor updates.
Target Stores (2013)	Third-Party Pivoting/POS Malware	Attackers hijacked valid access accounts belonging to an HVAC partner to penetrate the internal infrastructure, planting specific scraping malware directly on physical point-of-sale terminals.	Exfiltrated approximately 40 million financial credit profiles during live swipes, triggering customer attrition and widespread merchant reassessments.
Sony Pictures (2014)	Destructive Wiper Infrastructure Attack	A hostile group targeted corporate entities using aggressive wiper programs to delete server clusters, while concurrently leaking proprietary email records and unreleased intellectual properties.	Severely disrupted structural operational capacities, leaked corporate trade secrets.

CIA Triad Triangulation

Incident	Violated Vector	Analysis & Architectural Rationale
Equifax	Confidentiality	Defensive perimeters failed to safeguard database confidentiality. Unauthorized agents mapped internal directories to read and move restricted data out of corporate custody without clearance.
Colonial Pipeline	Availability	The cryptography mechanism rendered core logistical files inaccessible. Systems could not serve the infrastructure, resulting in a denial of service across functional operating units.
SolarWinds	Integrity	The build process was modified maliciously prior to distribution. Code validation chains were altered, stripping end-users of their ability to run untouched, trusted developer iterations.
Target Stores	Confidentiality	Payment metrics were scraped during transit across unencrypted memory streams. Live transaction data was read by foreign processes, breaking data privacy rules.
Sony Pictures	Confidentiality & Availability	Private communications were made public, destroying operational secrecy. Concurrently, data-wiping modules compromised system availability by bricking active servers.

3. Cybersecurity Infrastructure Tools

Defensive and offensive engineers use specific technical toolsets to test, audit, and validate structural compliance across corporate networks.

Tool Identity	Core Purpose	Operational Function	Practical Use Case Example
Wireshark	Network Traffic Inspection	Intercepts live data flows moving through local interface hardware, decoding individual packet headers and layer payloads into structured views.	When analyzing a host suspected of data exfiltration, an engineer runs Wireshark to verify outbound socket connections and cleartext transfers.
Nmap (Network Mapper)	Reconnaissance & Auditing	Probes target ip ranges via customizable layer packets to determine active nodes, open system ports, and running application versions.	Prior to deployment, engineers execute network discovery scans to uncover rogue server configurations and check if any high-risk ports are exposed.
Metasploit Framework	Penetration Testing & Exploit Verification	Aggregates an extensive library of verified vulnerabilities and modular code modules to launch safe verification exploits against target systems.	If a vulnerability scan highlights an outdated system component, an analyst can use Metasploit to safely test if the exploit is live and confirm remediation priorities.

4. Secure Home Network Topology

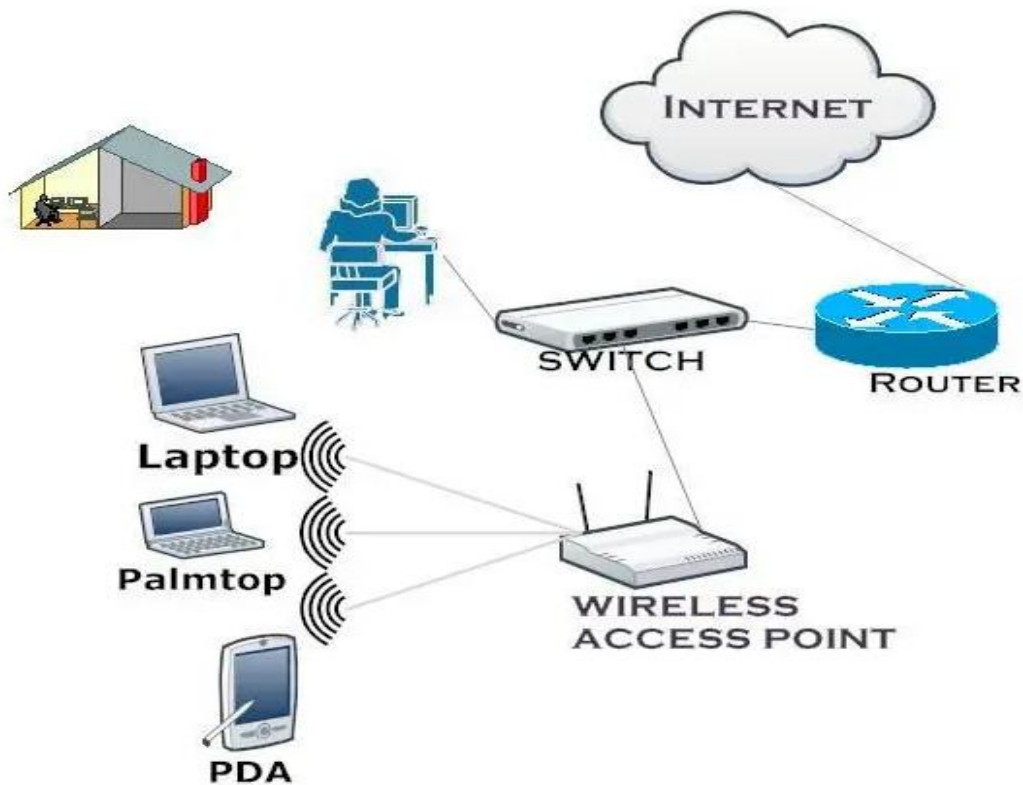


Figure 4.1 illustrates a basic home network topology where the **Internet** connects to a **router**, which distributes the network through a **switch** and a **wireless access point (WAP)**. The wireless access point provides Wi-Fi connectivity to devices such as a **laptop**, **palmtop**, and **PDA**, allowing them to access the internet and communicate within the local network.

5. Operational Profile: Daily Responsibilities of a Cybersecurity Analyst

A professional Cybersecurity Analyst manages the day-to-day balance of automated network perimeter parsing, host vulnerability reduction, and quick technical mitigation protocols. Rather than simply watching automated scripts or running simple command sequences, an analyst applies analytical tracking techniques to continuously observe networks and protect corporate digital boundaries against persistent threat entities.

The morning shift typically centers on reviewing SIEM (Security Information and Event Management) platforms. These collection engines organize log histories generated by distributed system elements, firewalls, and cloud access directories. An analyst filters through a large volume of non-threatening data points—such as legitimate user authentication errors or routine system configurations—to accurately locate high-value security events. The triage process demands sharp attention to isolate actual signs of security compromise, like unusual brute-force attempts or massive server requests originating from anomalous external IP blocks.

When a real event bypasses automated controls, the analyst handles the immediate investigative tasks. They analyze raw data across server platforms, verify domain registry accounts, and trace host endpoint activities to build a detailed incident timeline. This work aims to identify the attack's root entry point, confirm if assets were accessed, and determine whether malware has moved laterally across internal networks. The analyst must work carefully to confirm actual threats while ruling out benign errors before escalating to critical defense protocols.

In addition to active event tracking, analysts dedicate a major part of their shift to system surface reduction and structural security upkeep. This involves managing scheduled vulnerability scanners to detect unpatched software versions, documenting incident post-mortems, and updating firewall access tables to handle fresh threat vectors. Security analysts also communicate their findings across different business departments, explaining complex risk exposures to administrative managers and providing system teams with clear patching workflows based on the Common Vulnerability Scoring System (CVSS).